Bill of Materials (VBF-T5R1SINGLEMODEL-BOM-001)

Sheet: BOM

Component	Basis (formula)	g / cell	kg / kWh	Source note
Positive electrode active material (NMC811)	coating th x area x AM wt% x density	14.5881	0.794	AM/binder/additive split = literature default 96/2/2 wt% (base set has no binder/additive parameters)
Positive electrode conductive additive	literature default 2 wt% of electrode solids	0.3039	0.0165	literature default, annotated
Positive electrode binder (PVDF-class)	literature default 2 wt% of electrode solids	0.3039	0.0165	literature default, annotated
Negative electrode active material (graphite)	coating th x area x AM wt% x density	10.8567	0.5909	split = literature default 96/2/2 wt%
Negative electrode conductive additive	literature default 2 wt% of electrode solids	0.2262	0.0123	literature default, annotated
Negative electrode binder (SBR/CMC-class)	literature default 2 wt% of electrode solids	0.2262	0.0123	literature default, annotated
Separator	th x area x (1-por) x density	0.2161	0.0118	calc-energy layer caliber
Electrolyte	pore volume x 1.2 g/cm3 (lit.) x fill factor 1.0	6.616	0.3601	electrolyte density = literature value, annotated
Positive current collector (Al)	th x area x density	2.2183	0.1207	calc-energy layer caliber
Negative current collector (Cu)	th x area x density	5.5212	0.3005	calc-energy layer caliber
Enclosure + tabs	-	Not modeled	Not modeled	Not modeled
Total (contract caliber, electrolyte excluded)	sum of modeled layers	34.4606	1.8755	cell energy 18.3736 Wh from r5_final_archN_energy_dfn.json
Total (incl. electrolyte, literature density)	contract + electrolyte	41.0766	2.2356	electrolyte mass at 1.2 g/cm3 literature density; electrolyte excluded from contract ED caliber